

Chapter 4: Data Link Layer

The data link layer has a number of specific functions it can carry out. These functions include

- Provide a well-defined service interface to the network layer.
- Deal with transmission errors.
- Regulate the flow of data so that fast senders do not swamp slow receivers.

Services Provided to the Network Layer

The function of the data link layer is to provide services to the network layer.

The actual services offered can vary from system to system. Three reasonable possibilities that are we will consider in turn are:

- Unacknowledged connectionless service
- Acknowledged connectionless service
- Acknowledged connection-oriented service

Data Link layer is divided into two sub layers:

- Logical Link Control (LLC)
- Medium Access Control (MAC)

Logical Link Control (LLC)

LCC is responsible for error control & flow control. It is defined by IEEE 802.2 Standard. The LLC sub-layer acts as an interface between the media access control (MAC) sub-layer and the network layer

Medium Access Control (MAC)

MAC is responsible for multiple access resolutions. It provides addressing and channel access control mechanisms. It also provides End Devices Addressing Mechanism using physical address

Physical (MAC) addressing Overview

It is also called Physical Address OR Hardware Address. It is 48-bit address represented in Hexadecimal number. Example:

01:23:45:67:89:ab

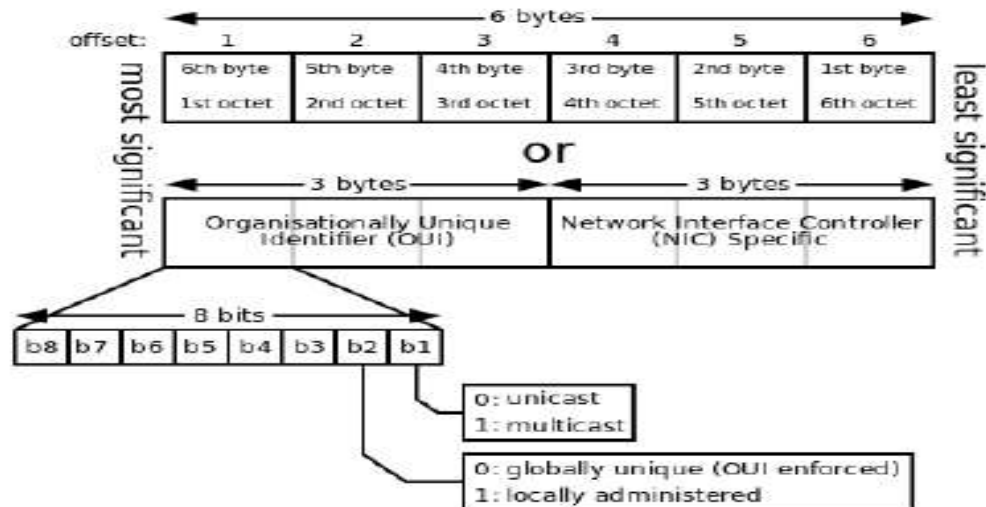


Figure: MAC Address

Upper 3 bytes represents the OUI (Organization Unique Identifier) also called Manufacturer ID and lower 3 bytes represent the Device ID

Framing

Data link layer translates the physical layer's raw bit stream into discrete units called *frames* and encapsulates a network layer datagram into frame. The Process of creating Frames by the Data Link Layer is known as Framing.

One of the method of framing is fixed size Framing that have Fixed Length and no need to define boundaries for Frames. Example, ATM Frames (54 byte cells)

Other method of framing is variable size framing. Its types are:

- Character count
- Flag bytes with byte stuffing
- Starting and ending flags, with bit stuffing.
- Physical layer coding violations

Character Count

The first framing method uses a field in the header to specify the number of bytes in the frame. When the data link layer at the destination sees the byte count, it knows how many bytes follow and hence where the end of the frame is. This technique is shown in figure for four small example frames of sizes 5, 5, 8, and 8 bytes, respectively.

The trouble with this algorithm is that the count can be garbled by a transmission error. For example, if the byte count of 5 in the second frame of figure becomes a 7 due to a single bit flip,

the destination will get out of synchronization. It will then be unable to locate the correct start of the next frame. Even if the checksum is incorrect so the destination knows that the frame is bad, it still has no way of telling where the next frame starts.

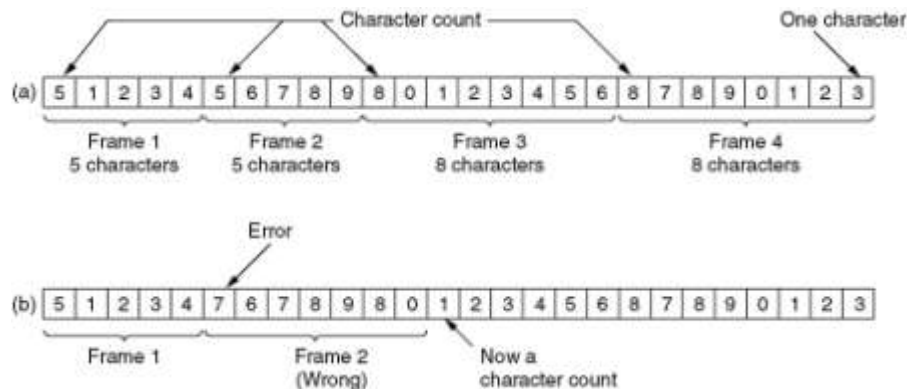


Figure: A byte stream a) Without errors b) With one error

Flag bytes with byte stuffing

Often the same byte, called a **flag byte**, is used as both the starting and ending delimiter. This byte is shown in Figure as FLAG. Two consecutive flag bytes indicate the end of one frame and the start of the next. Thus, if the receiver ever loses synchronization it can just search for two flag bytes to find the end of the current frame and the start of the next frame.

However, there is still a problem we have to solve. It may happen that the flag byte occurs in the data. One way to solve this problem is to have the sender's data link layer insert a special escape byte (ESC) just before each "accidental" flag byte in the data. The data link layer on the receiving end removes the escape bytes before giving the data to the network layer. This technique is called **byte stuffing**.

Of course, the next question is: what happens if an escape byte occurs in the middle of the data? The answer is that it, too, is stuffed with an escape byte. At the receiver, the first escape byte is removed, leaving the data byte that follows it. Some examples are shown in Figure.

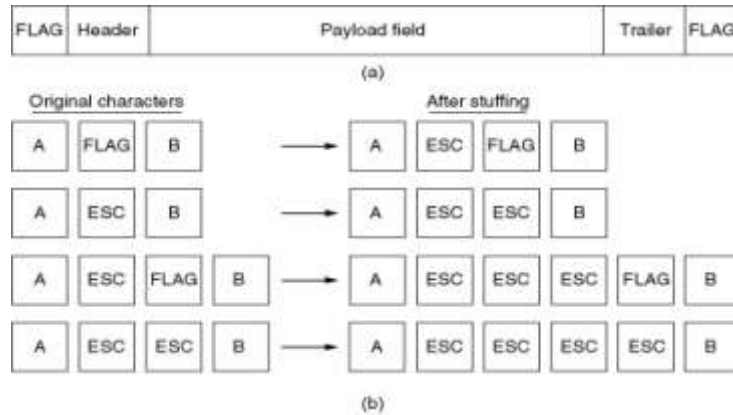


Figure : a) A frame delimited by flag bytes. b) Four examples of byte sequences before and after byte stuffing

Starting and ending flags, with bit stuffing

The new technique allows data frames to contain an arbitrary number of bits and allows character codes with an arbitrary number of bits per character. It works like this. Each frame begins and ends with a special bit pattern, 01111110 (in fact, a flag byte). Whenever the sender's data link layer encounters five consecutive 1s in the data, it automatically stuffs a 0 bit into the outgoing bit stream. This bit stuffing is analogous to byte stuffing, in which an escape byte is stuffed into the outgoing character stream before a flag byte in the data.

When the receiver sees five consecutive incoming 1 bits, followed by a 0 bit, it automatically destuffs (i.e., deletes) the 0 bit. Just as byte stuffing is completely transparent to the network layer in both computers, so is bit stuffing. If the user data contain the flag pattern, 01111110, this flag is transmitted as 0111111010 but stored in the receiver's memory as 01111110. Figure gives an example of bit stuffing.

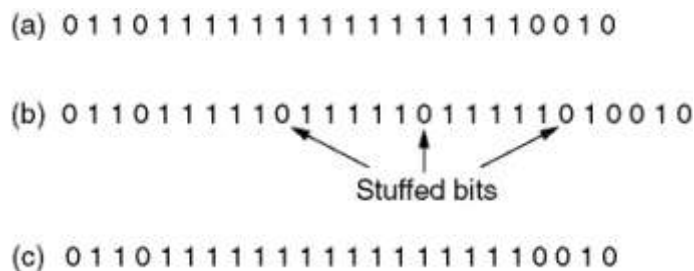


Figure: Bit stuffing. a) The original data b) The data as they appear on the line. c) The data as they are stored in the receiver's memory after de-stuffing.

Physical layer coding violations

The last method of framing is only applicable to networks in which the encoding on the physical medium contains some redundancy. For example, some LANs encode 1 bit of data by using 2 physical bits. Normally, a 1 bit is a high-low pair and a 0 bit is a low-high pair. The scheme means that every data bit has a transition in the middle, making it easy for the receiver to locate the bit boundaries. The combinations high-high and low-low are not used for data but are used for delimiting frames in some protocols.

Flow Control

Another important design issue that occurs in the data link layer (and higher layers as well) is what to do with a sender that systematically wants to transmit frames faster than the receiver can accept them. This situation can easily occur when the sender is running on a fast (or lightly loaded) computer and the receiver is running on a slow (or heavily loaded) machine. The sender keeps pumping the frames out at a high rate until the receiver is completely swamped.

Two approaches are commonly used. In the first one, feedback-based flow control, the receiver sends back information to the sender giving it permission to send more data or at least telling the sender how the receiver is doing. In the second one, rate-based flow control, the protocol has a built-in mechanism that limits the rate at which senders may transmit data, without using feedback from the receiver. Here, we will study feedback-based flow control schemes because rate-based schemes are never used in the data link layer.

Stop and Wait protocol

If data frames arrive at the receiver site faster than they can be processed, the frames must be stored until their use. Normally, the receiver does not have enough storage space, especially if it is receiving data from many sources. This may result in either the discarding of frames or denial of service. To prevent this, we somehow need to tell the sender to slow down i.e., stop to transmit and wait for receiver acknowledgement signals

Stop and Wait: Normal Operation

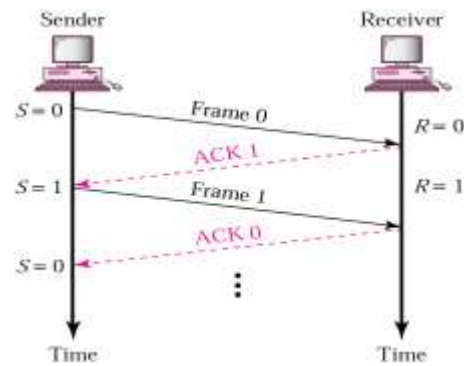


Figure (a): Stop and Wait: Normal Operation

Sender keeps a copy of the last frame until it receives an acknowledgement. For identification, both data frames and acknowledgements (ACK) frames are numbered alternatively 0 and 1. Sender has a control variable (S) that holds the number of the recently sent frame. (0 or 1) Receiver has a control variable (R) that holds the number of the next frame expected (0 or 1). Sender starts a timer when it sends a frame. If an ACK is not received within a allocated time period, the sender assumes that the frame was lost or damaged and resends it. Receiver send only positive ACK if the frame is intact. ACK number always defines the number of the next expected frame.

Stop-and-Wait, lost frame

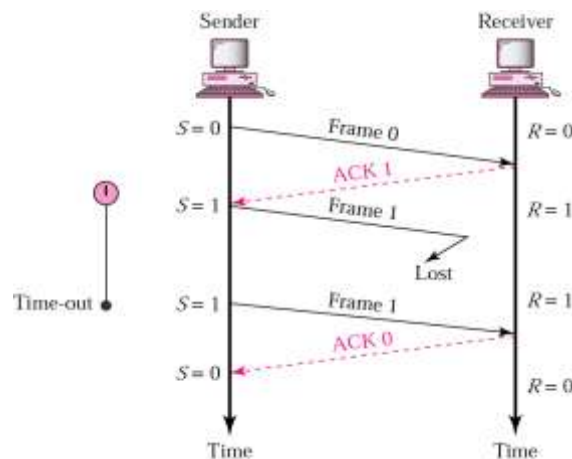


Figure (b): Stop-and-Wait ARQ, lost frame

When a receiver receives a damaged frame, it discards it and keeps its value of R. After the timer at the sender expires, another copy of frame 1 is sent.

Stop and wait ,Lost ACK

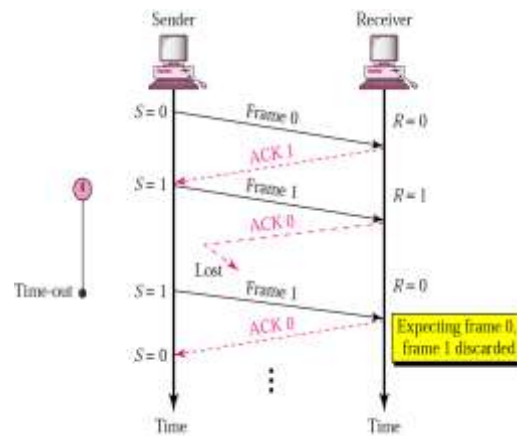


Figure (c): Stop and wait ,Lost ACK

If the sender receives a damaged ACK, it discards it. When the timer of the sender expires, the sender retransmits frame 1. Receiver has already received frame 1 and expecting to receive frame 0 ($R=0$). Therefore it discards the second copy of frame 1.

Stop-and-Wait, delayed ACK

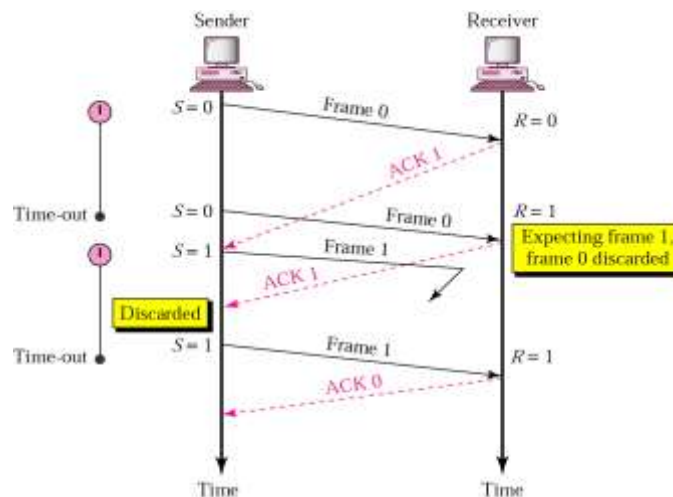


Figure (d): Stop-and-Wait ARQ, delayed ACK

The ACK can be delayed at the receiver or due to some problem. It is received after the timer for frame 0 has expired. Sender retransmitted a copy of frame 0. However, $R = 1$ means receiver expects to see frame 1. Receiver discards the duplicate frame 0. Sender receives 2 ACKs, it discards the second ACK.

Disadvantage of Stop-and-Wait

In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged. This is not a good use of transmission medium. To improve efficiency, multiple frames should be in transition while waiting for ACK.

A Protocol Using Go-Back-N

Allowing the sender to transmit up to w frames before blocking, instead of just 1. With a large enough choice of w the sender will be able to continuously transmit frames since the acknowledgements will arrive for previous frames before the window becomes full, preventing the sender from blocking.

To find an appropriate value for w we need to know how many frames can fit inside the channel as they propagate from sender to receiver. This procedure requires additional features to be added to Stop-and-Wait ARQ. It uses Sequence Numbering Techniques to track the Frames. It can send one cumulative acknowledgment for several frames.

Sequence Number

- Frames from a sender are numbered sequentially
- We need to set a limit since we need to include the sequence number of each frame in the header
- If the header of the frame allows m bits for sequence number, the sequence numbers range from 0 to $2^m - 1$.
- for $m = 3$, sequence numbers are: 0, 1, 2, 3, 4, 5, 6, 7.
- We can repeat the sequence number.
- Sequence numbers are:
 - 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, ...
- Sliding window define the range of Sequences Number
- Here Sender Sliding window define the window size=7
- Total Number of Frames that can be sent without receiving ACKs is 7

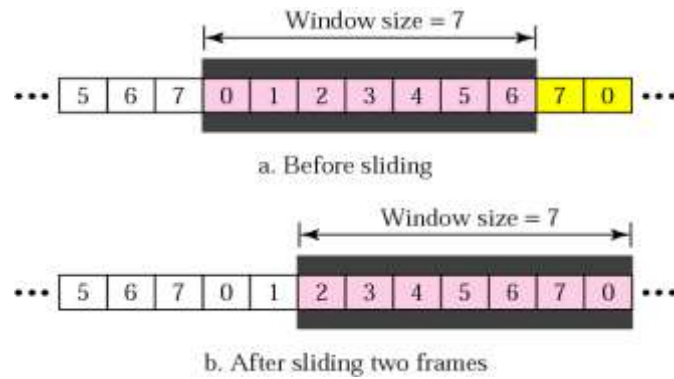


Figure: Go-Back-N ARQ: Sender sliding window

Control Variables

- Sender has 3 variables: S, SF, and SL
- S holds the sequence number of recently sent frame
- SF holds the sequence number of the first frame
- SL holds the sequence number of the last frame
- Receiver only has the one variable, R, that holds the sequence number of the frame it expects to receive. If the seq. no. is the same as the value of R, the frame is accepted, otherwise rejected.

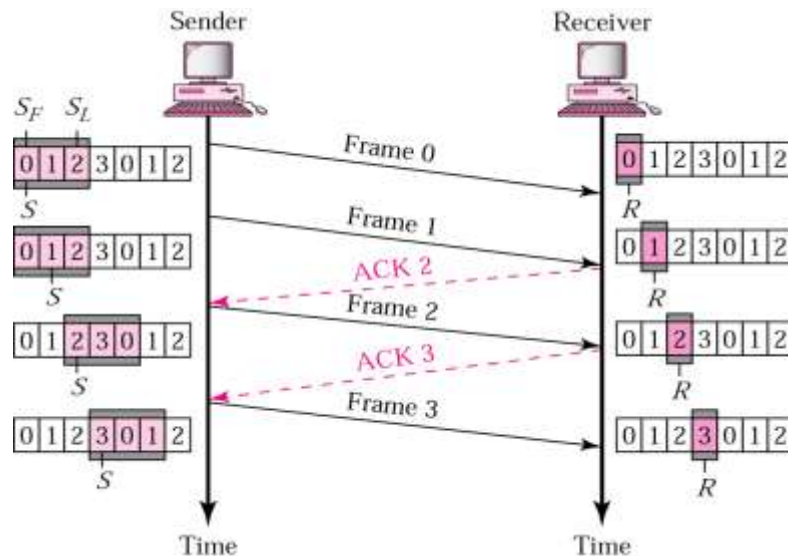


Figure: Go-Back-N ARQ : Normal Operation

Selective Repeat

The other general strategy for handling errors when frames are pipelined is called **selective repeat**. When it is used, a bad frame that is received is discarded, but any good frames received after it are accepted and buffered. When the sender times out, only the oldest unacknowledged frame is retransmitted. If that frame arrives correctly, the receiver can deliver to the network layer, in sequence, all the frames it has buffered. Selective repeat corresponds to a receiver window larger than 1. This approach can require large amounts of data link layer memory if the window is large.

Selective repeat is often combined with having the receiver send a negative acknowledgement (NAK) when it detects an error, for example, when it receives a checksum error or a frame out of sequence. NAKs stimulate retransmission before the corresponding timer expires and thus improve performance.

It is more efficient for noisy links. The Selective Repeat Protocol also uses two windows: a send window and a receive window. Size of the Send window and Receive window are same.

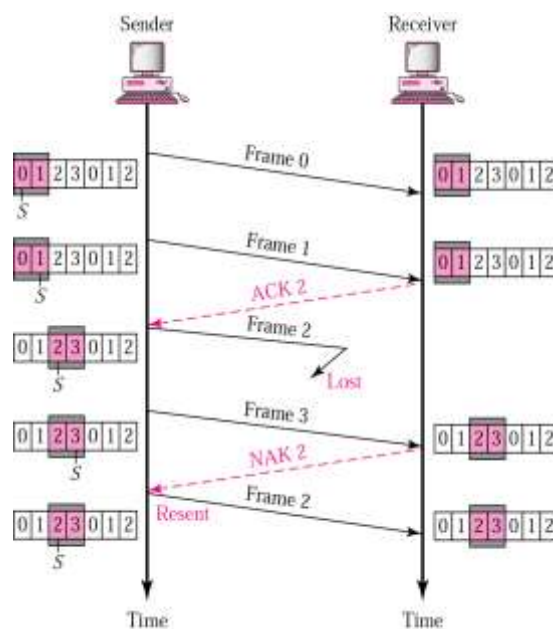


Figure: Selective Repeat Request, lost frame

What happens when we get a bad frame?

- **Go back N** – Ask the sender to go back and start retransmitting from the lost frame.
- **Selective repeat** – Ask the sender to repeat the particular frames that were lost.

Error Control

Suppose that the sender just kept outputting frames without regard to whether they were arriving properly. This might be fine for unacknowledged connectionless service, but would most certainly not be fine for reliable, connection-oriented service.

Typically, the protocol calls for the receiver to send back special control frames bearing positive or negative acknowledgements about the incoming frames. If the sender receives a positive acknowledgement, it knows the frame has arrived safely and, a negative acknowledgement means that something has gone wrong, and the frame must be transmitted again.

An additional complication comes from the possibility that hardware troubles may cause a frame to vanish completely (e.g., in a noise burst). This possibility is dealt with by introducing timers into the data link layer. The timer is set to expire after an interval long enough for the frame to reach the destination, be processed there, and have the acknowledgement propagate back to the sender.

However, if either the frame or the acknowledgement is lost, the timer will go off, alerting the sender to a potential problem. The obvious solution is to just transmit the frame again. However, when frames may be transmitted multiple times there is a danger that the receiver will accept the same frame two or more times and pass it to the network layer more than once. To prevent this from happening, it is generally necessary to assign sequence numbers to outgoing frames, so that the receiver can distinguish retransmissions from originals.

Error control includes both error detection and error correction.

Error Detection

- CRC (Cyclic Redundancy Check)
- Parity Check
- CheckSum

CRC (Cyclic Redundancy Check)

Given a k -bit frame or message, the transmitter generates an n -bit sequence, known as a frame check sequence (FCS), so that the resulting frame, consisting of $(k+n)$ bits, is exactly divisible by some predetermined number. The receiver then divides the incoming frame by the same number and, if there is no remainder, assumes that there was no error.

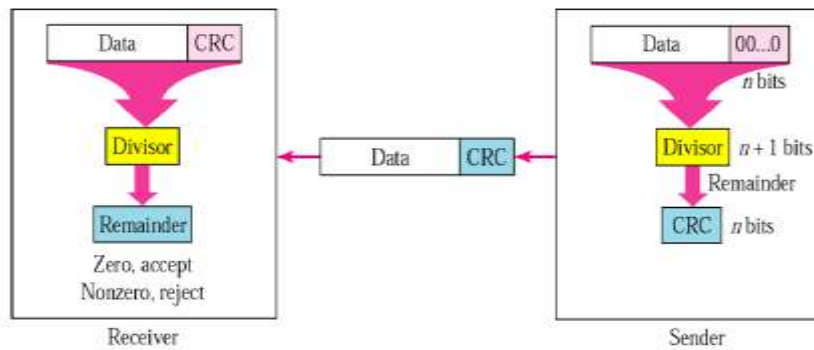


Figure (a): CRC

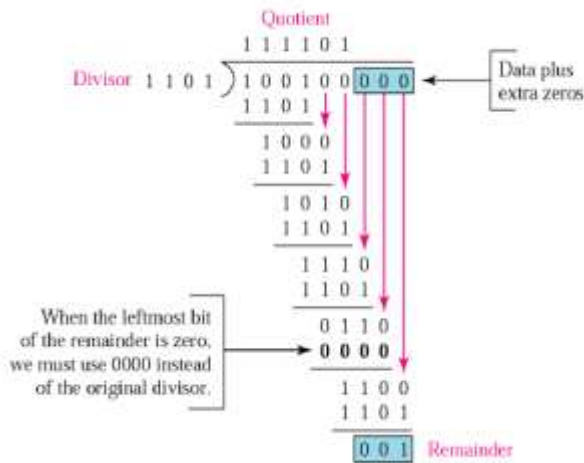


Figure (b): CRC Encoding

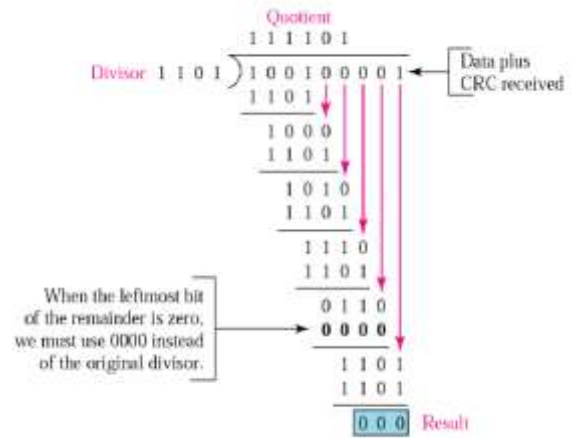


Figure (c): CRC Decoding

Parity Check

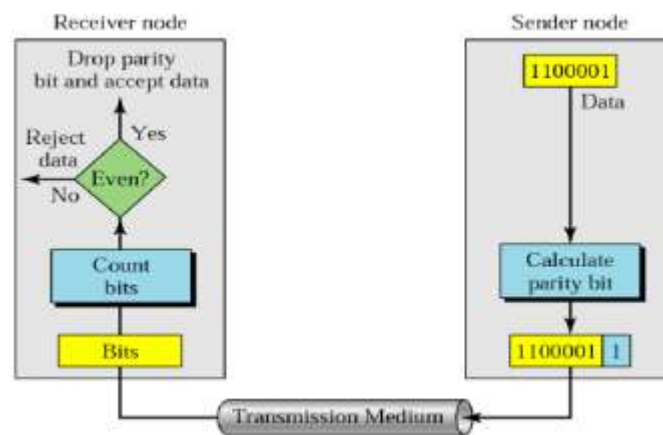


Figure (a): Parity Check (Even Parity)

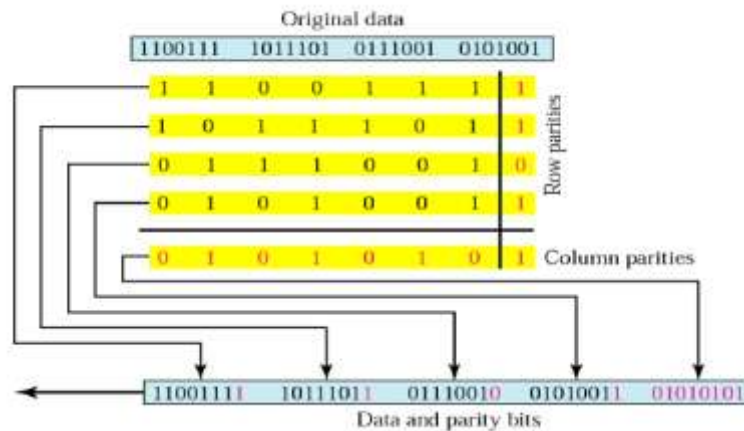


Figure (b): Two dimensional Parity check

Error Correction

Forward Error Control (FEC)

Each block transmitted contains extra information which can be used to detect the presence of errors and determine the position in the bit stream of the errors.

Backward (Feedback) Error Control (BEC)

Extra information is sufficient to only detect the presence of errors. If necessary, a retransmission control scheme is used to request that another copy of the erroneous information be sent.

Hamming codes

Given any two code words that may be transmitted or received—say, 10001001 and 10110001. It is possible to determine how many corresponding bits differ. In this case, 3 bits differ.

For example: 10001001 and 10110001 differs in third, fourth and fifth bit position.

The number of bit positions in which two code words differ is called the **Hamming Distance (d)**

Example:

$W1=10001001$, $W2=10110001$, then $d(W1, W2) = 3$

The minimum Hamming distance (or “minimum distance”) of the scheme is the smallest number of bit errors that changes one valid codeword into another.

This scheme can detect any combination of $\leq D-1$ bit errors and correct any combination of strictly less than $D/2$ bit errors.

Suppose only 4 valid code words are:

00000 00000, 00000 11111, 11111 00000, and 11111 11111

Minimum distance is $D=5$, so any combination of ≤ 4 bit errors can be detected and any combination of ≤ 2 bit errors can be corrected, but 3 bit errors can't be properly corrected.

If 00000 00000 transmitted, $W=00000\ 00011$ received: $d(W, C1)=2$, $d(W, C2)=3$, $d(W, C3)=7$, and $d(W, C4)=8$ receiver takes $C1=00000\ 00000$ as transmitted codeword (errors corrected)

If 00000 00000 transmitted, $X=00000\ 00111$ received: $d(X, C1)=3$, $d(X, C2)=2$, $d(X, C3)=8$, and $d(X, C4)=7$ receiver takes $C2=00000\ 11111$ as transmitted codeword (in this case, error correction fails)

Random Access Protocols

In this method, there is no control station. Any station can send the data. There is no scheduled time for a stations to transmit. They can transmit in random order. The various random access methods are:

- ALOHA
- CSMA (Carrier Sense Multiple Access)
- CSMA/CD (Carrier Sense Multiple Access with Collision Detection)
- CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

ALOHA

Any terminal is allowed to transmit without considering whether channel is idle or busy. If packet is received correctly, the base station transmits an acknowledgement. If no acknowledgement is received, it assumes the packet to be lost and it retransmits the packet after waiting a *random time*. There are two different versions of ALOHA:

- Pure ALOHA
- Slotted ALOHA

Pure ALOHA

In pure ALOHA, stations transmit frames whenever they have data to send. When two stations transmit simultaneously, there is collision and frames are lost. In pure ALOHA, whenever any station transmits a frame, it expects an acknowledgement from the receiver. If acknowledgement is not received within specified time, the station assumes that the frame has been lost. If the

frame is lost, station waits for a random amount of time and sends it again. This waiting time must be random, otherwise, same frames will collide again and again. Whenever two frames try to occupy the channel at the same time, there will be collision and both the frames will be lost. If first bit of a new frame overlaps with the last bit of a frame almost finished, both frames will be lost and both will have to be retransmitted.

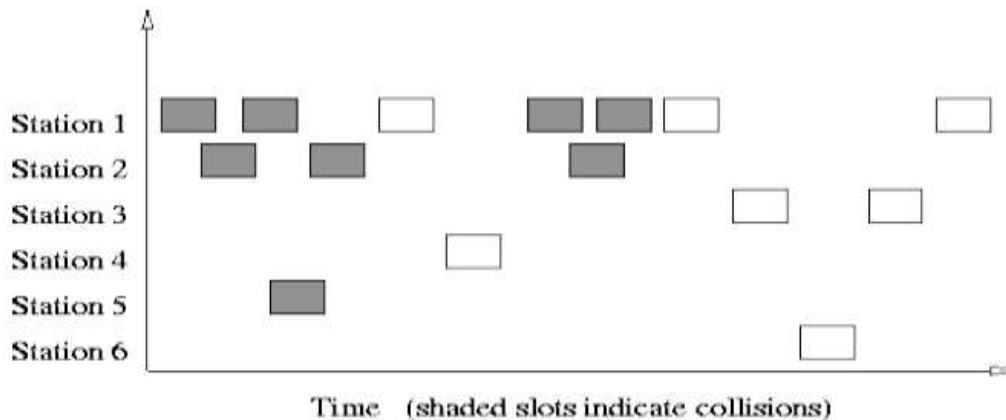


Figure: Pure ALOHA

Slotted ALOHA

Slotted ALOHA was invented to improve the efficiency of pure ALOHA. In slotted ALOHA, time of the channel is divided into intervals called slots. The station can send a frame only at the beginning of the slot and only one frame is sent in each slot. If any station is not able to place the frame onto the channel at the beginning of the slot, it has to wait until the next time slot. There is still a possibility of collision if two stations try to send at the beginning of the same time slot.

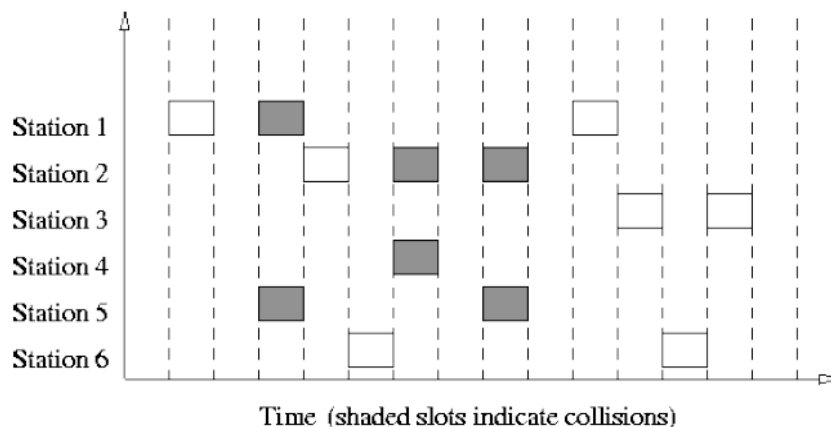


Figure: Slotted ALOHA

CSMA

CSMA was developed to overcome the problems of ALOHA i.e. to minimize the chances of collision. It is based on the principle "sense before transmit" or "listen before talk." In CSMA, node verifies the absence of other traffic before transmitting on a shared transmission medium. Multiple access means that multiple stations send and receive on the medium. Each station first listen to the medium before Sending.

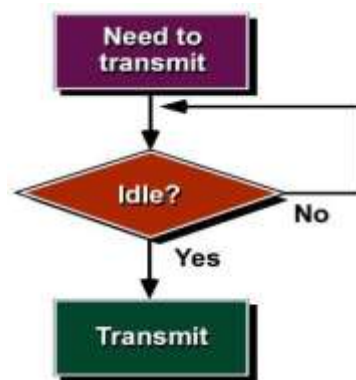


Figure (a): CSMA

The chances of collision still exist because of propagation delay. There are three different types of CSMA protocols:

- 1-Persistent CSMA
- Non-Persistent CSMA
- P-Persistent CSMA

1-Persistent CSMA

In this method, station that wants to transmit data, continuously senses the channel to check whether the channel is idle or busy. If the channel is busy, station waits until it becomes idle. When the station detects an idle channel, it immediately transmits the frame. This method has the highest chance of collision because two or more stations may find channel to be idle at the same time and transmit their frames.

Non-Persistent CSMA

A station that has a frame to send senses the channel. If the channel is idle, it sends immediately. If the channel is busy, it waits a random amount of time and then senses the channel again. It reduces the chance of collision because the stations wait for a random amount of time. It is unlikely that two or more stations will wait for the same amount of time and will retransmit at the same time.

P-Persistent CSMA

In this method, the channel has time slots such that the time slot duration is equal to or greater than the maximum propagation delay time. When a station is ready to send, it senses the channel. If the channel is busy, station waits until next slot. If the channel is idle, it transmits the frame.

CSMA/CD (CSMA with Collision Detection)

In this protocol, the station senses the channel before transmitting the frame. If the channel is busy, the station waits. Additional feature in CSMA/CD is that the stations can detect collisions. The stations abort their transmission as soon as they detect collision. In CSMA/CD, the station that sends its data on the channel, continues to sense the channel even after data transmission. If collision is detected, the station aborts its transmission and waits for a random amount of time & sends its data again. As soon as a collision is detected, the transmitting station release a ***jam signal***. Jam signal alerts other stations. Stations are not supposed to transmit immediately after the collision has occurred.

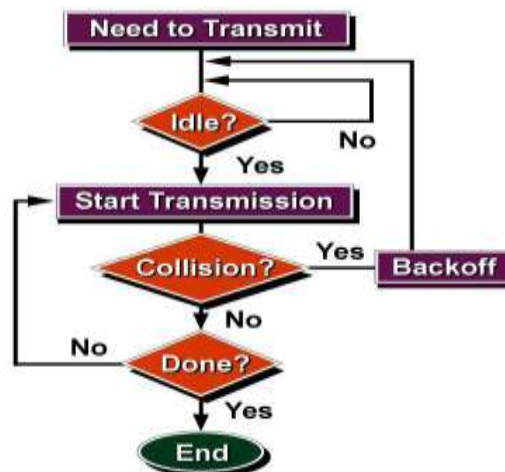


Figure: CSMA/CD

IEEE 802.3: Ethernet

The IEEE standard for Ethernet is 802.3. Ethernet operates in two areas of the OSI model: the lower half of the data link layer, which is known as the MAC sub layer, and the physical layer.

The CSMA/CD is the access method used in Ethernet to detect and avoid collision in network. Beside carrier sensing, collision detection and the binary exponential back-off algorithm, the

standard also describes the format of the frames and the type of encoding used for transmitting frames. The minimum length of frames can be varied from network to network.

The most commonly installed Ethernet systems are called 10BASE-T and provide transmission speeds up to 10 Mbps. Devices are connected to the cable and compete for access using a Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol. **Fast Ethernet** or 100BASE-T provides transmission speeds up to 100 megabits per second and is typically used for LAN backbone systems, supporting workstations with 10BASE-T cards. **Gigabit Ethernet** provides an even higher level of backbone support at 1000 megabits per second (1 gigabit or 1 billion bits per second). 10-Gigabit Ethernet provides up to 10 billion bits per second.

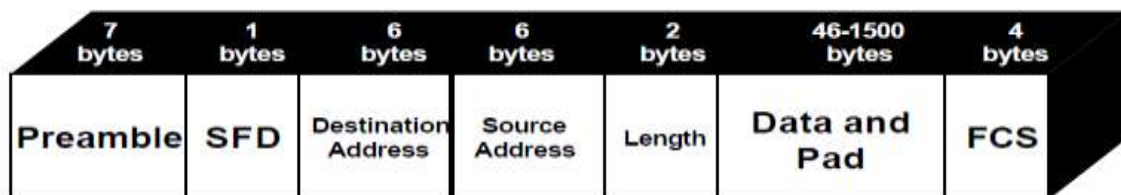


Figure: Ethernet Frame Format

IEEE 802.4: Token Bus

802.3 suffer from the difficulty of large delay in getting the access and at the same time poor performance under heavy load. There are also no priorities in 802.3, making them unsuited for real time systems. Token passing protocols were proposed and were found to be very attractive for situations with heavy load. The basic idea is to generate a token in the network. Only the holder of the token can transmit. Thus with one token, only one station can transmit at a time, eliminating collisions totally. Normally a token can be held by a user for a prescribed time only after which it has to be passed to the next station. If the user finishes his transmission before his token holding time is over, he passes the token to the next user.

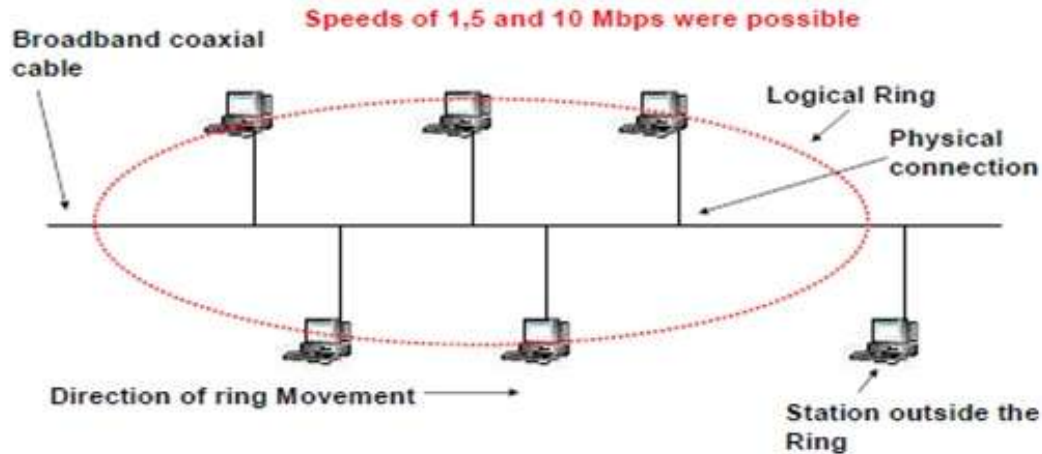


Figure (a): Token bus layout

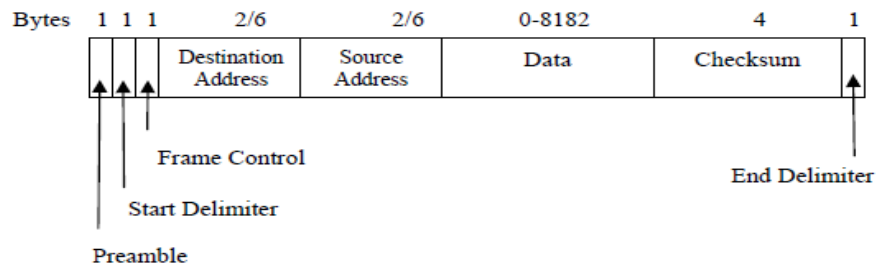


Figure (b): IEEE 802.4 Frame format

IEEE 802.5: Token Ring

Ring is not a broadcast medium but a collection of point-to-point links forming a circle. Rings can be based on twisted pair, coaxial or a fiber optics cable. Channel access problem is solved with the help of a special frame called a “Token”. A free token circulates the ring when all stations are idle. A station wishing to transmit must wait until it detects a free token passing by. It then seizes the token by changing the token bit to transform it into the start-of-frame sequence for a data frame. The data to be transmitted is then appended. The frame on the ring will make a round trip and then removed by the transmitting station.

Use of wire centers

Cable breaks can lead to ring failure. This problem can be resolved with the help of a Wire Center. A wire center has bypass relays which draw current from the station. If a station is powered down the relays close thereby removing the station from the ring and maintaining the

ring. Relays can be operated by software for network management. The wire centers make the ring a star-shaped ring.

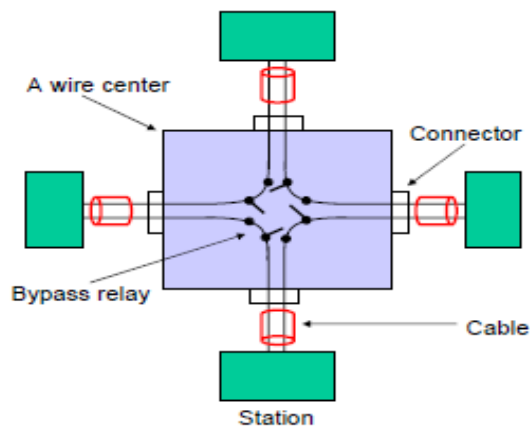


Figure (a): Four stations connected to a wire center

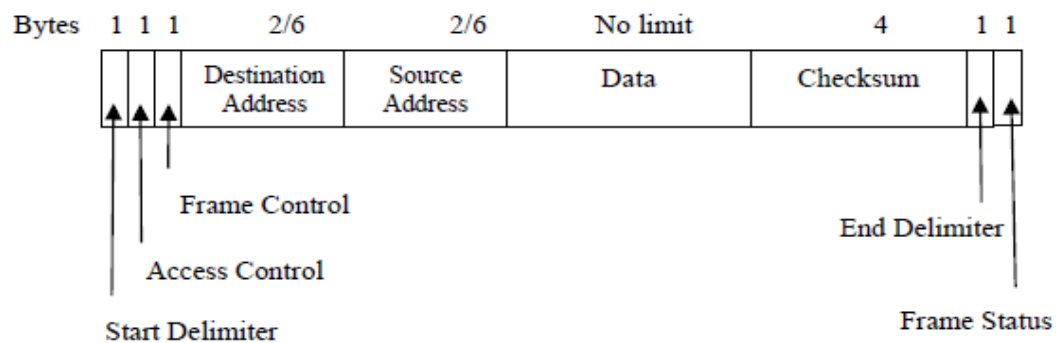
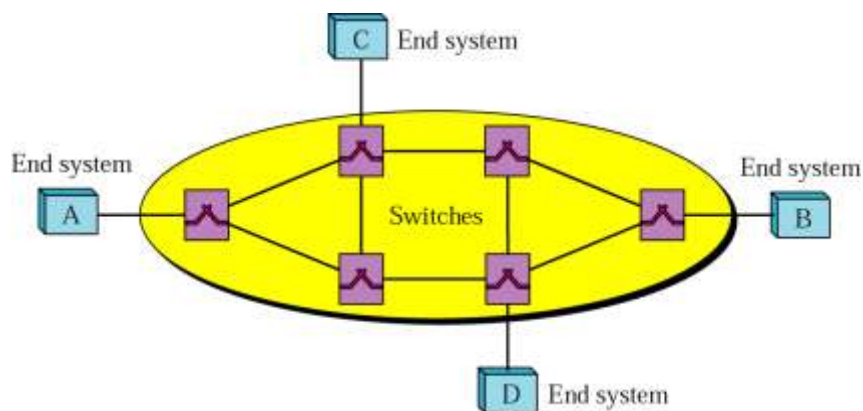
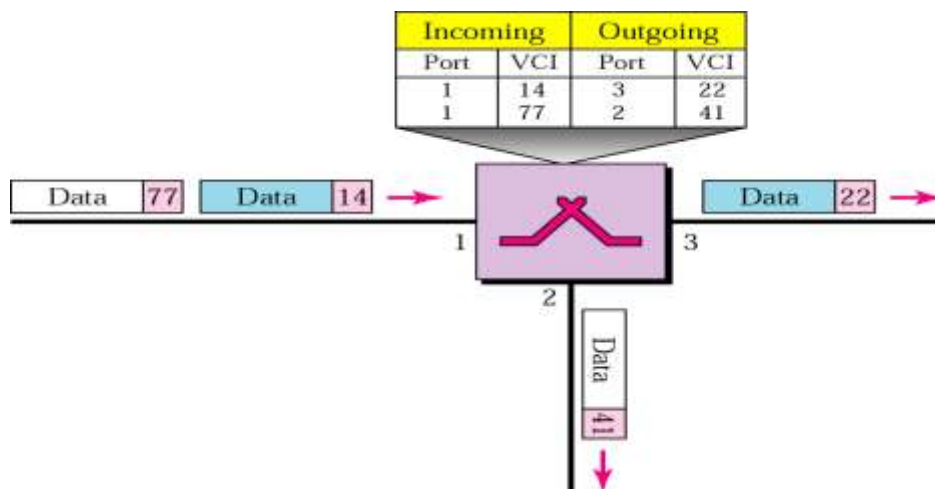
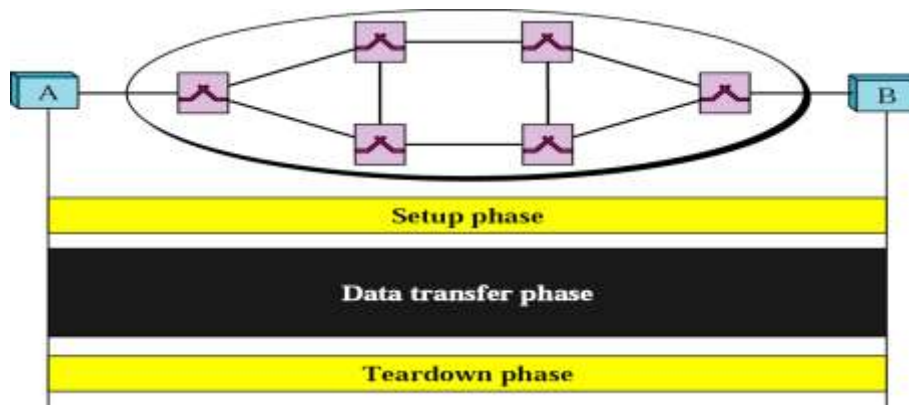
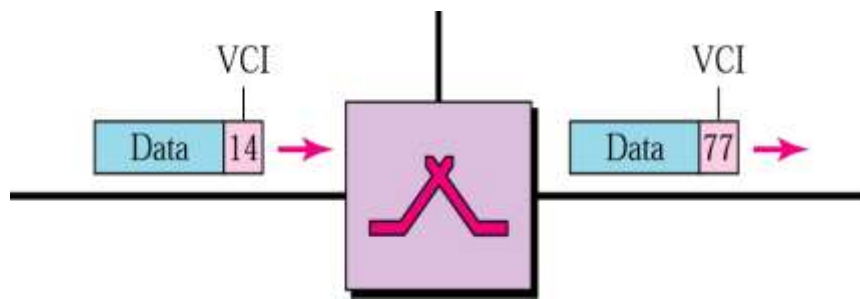
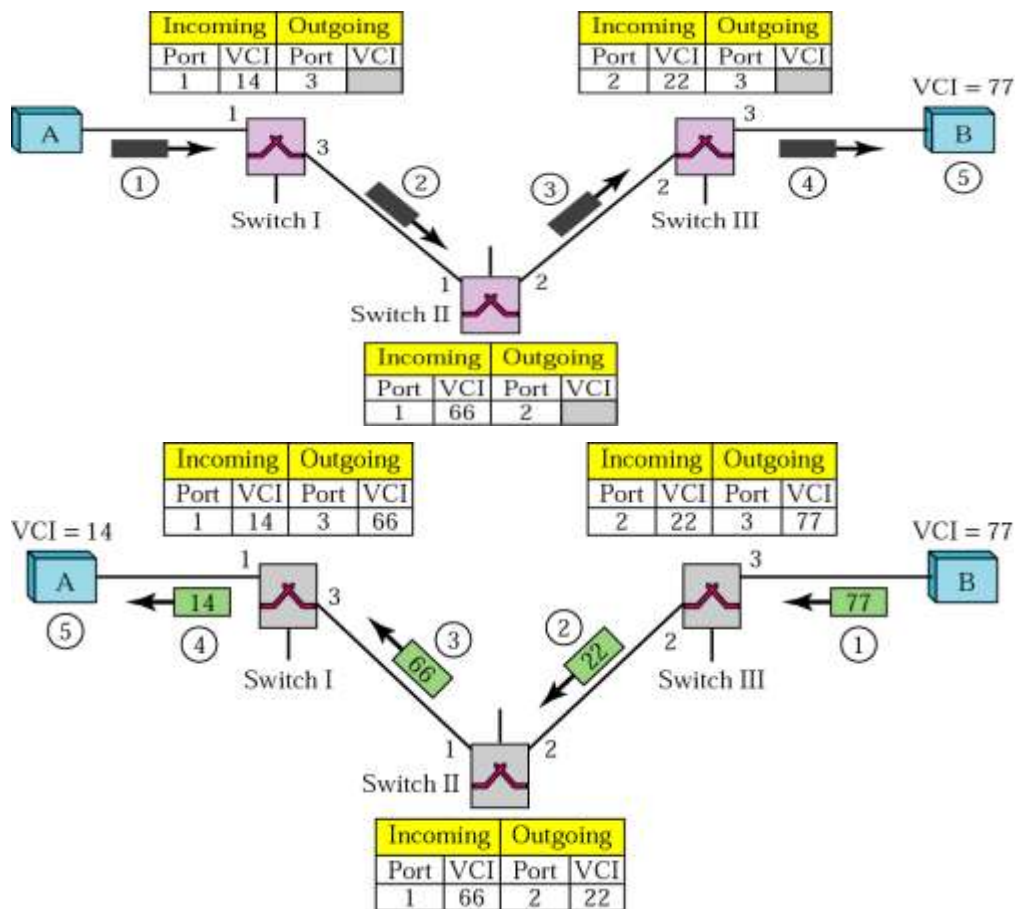
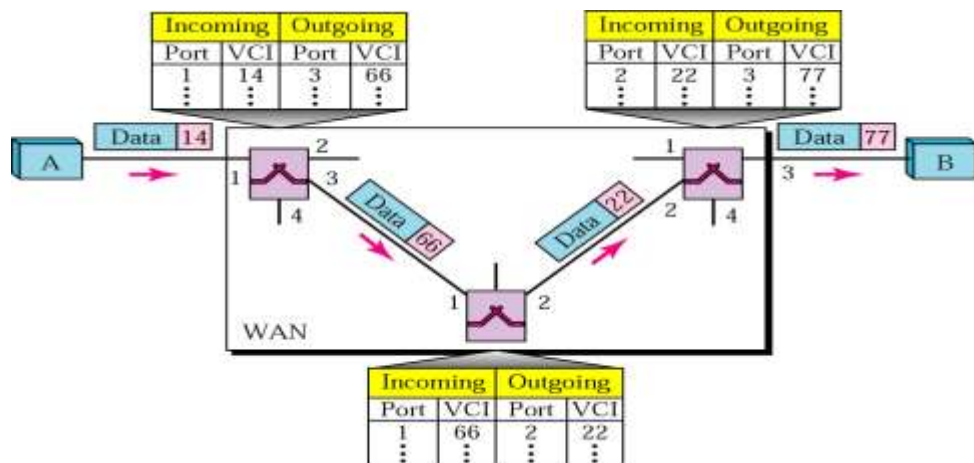


Figure (b): IEEE 802.5 Frame Format

Virtual Circuit Switching







Frame Relay

Frame Relay is a virtual-circuit wide-area network. It is connection-oriented network with no error control and no flow control. Therefore, packets are always delivered in the same order they were sent. It does not have a retransmission policy if a frame is damaged which is simply dropped. Frame relay was designed in this way to provide fast transmission capability for more reliable media and for those protocols that have flow and error control at the higher layers. Like any other connection-oriented systems, Frame Relay involves three phases, namely, connection establishment phase, data transmission phase and connection termination phase. Frame Relay is used for interconnecting LANs at multiple company offices, and can also be used as a backbone network.

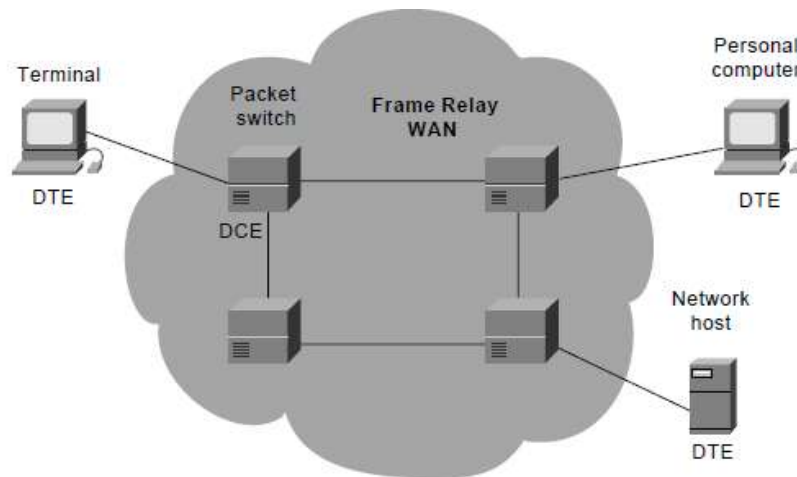


Figure: Basic frame relay setup

A Frame Relay network may be considered as a cloud that consists of switches, and customer nodes. The switch acts as Data Communication Equipment (DCE) and the customer equipment works as Data Terminal Equipment (DTE). A virtual circuit is established between the DTE and corresponding DCE. A virtual circuit is identified by a DLCI (Data Link Connection Identifier) number. On a given physical channel, there cannot be two DLCIs which are identical.

Permanent Virtual Circuit (PVC) and Switched Virtual Circuit (SVC)

In PVC, a permanent connection is established between two distant hosts that communicate frequently, and the process involves recording corresponding table entry for all the switches by the administrator. An outgoing DLCI is given to the source and an incoming DLCI is given to the destination. One of the drawbacks of PVC is that it is costly because the connection needs to

be kept connected all the time even when it is idle. Another drawback is that separate PVC is needed if a hosts needs to be connected to multiple hosts.

In a SVC, a temporary connection is established between the two distant hosts whenever data transfer is required, that is dynamically and is soon disconnected when the job is done. Thus SVC overcomes the shortcomings of the PVC. In an SVC three steps are required: (1) establish a connection, (2) transfer data and (3) terminate the connection.

Asynchronous Transfer Mode (ATM)

It is a switching technique used by telecommunication networks that uses asynchronous time-division multiplexing to encode data into small, fixed-sized cells. This is different from Ethernet or Internet, which use variable packet sizes for data or frames. *ATM* (also called *cell relay*) was originally designed to carry both voice and data traffic over WANs. ATM is the core protocol used over the synchronous optical network (SONET) backbone of the integrated digital services network (ISDN).

Characteristics of ATM

- ATM was designed with fixed cell structure in mind.
- ATM creates fixed routes between two points before data transfer begins
- ATM uses a mesh topology
 - This mesh is made up of point-to-point, full duplex circuits that interconnect ATM switches.
- ATM addressing uses *virtual channels (VCs)*.
- VCs can be set up in one of two ways:
 - Permanent Virtual Circuits (PVCs): permanent virtual circuits set up for long periods.
 - Switched Virtual Circuits (SVCs) : temporary virtual circuits set up for one transmission and deleted when the transmission is completed.

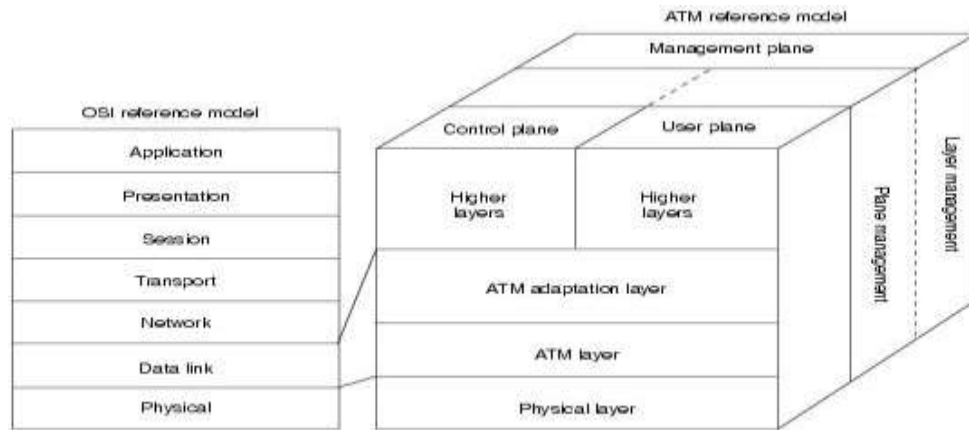


Figure: ATM

- ATM functionality corresponds to the physical layer and part of the data link layer of the OSI reference model.
- The ATM reference model, as shown in Figure, consists of the following planes:
 - Control: This plane is responsible for generating and managing signaling requests.
 - User: This plane is responsible for managing the transfer of data.
 - Management: This plane contains two components:
 - Layer management manages layer-specific functions, such as the detection of failures and protocol problems.
 - Plane management manages and coordinates functions related to the complete system.
- The ATM reference model consists of the following ATM layers:
 - Physical layer: Analogous to the physical layer of the OSI reference model, the main functions of the ATM physical layer are as follows:
 - Cells are converted into a bit stream
 - The transmission and receipt of bits on the physical medium are controlled
 - ATM layer: Combined with the ATM adaptation layer, the ATM layer is roughly analogous to the data link layer of the OSI reference model.
 - The ATM layer provides routing, traffic management, switching and multiplexing services.

- It processes outgoing traffic by accepting 48-byte segment from the AAL sub-layers and transforming them into 53-byte cell by addition of a 5-byte header.
- ATM adaptation layer (AAL): Combined with the ATM layer, the AAL is roughly analogous to the data link layer of the OSI model.
 - The AAL is responsible for isolating higher-layer protocols from the details of the ATM processes.
 - AAL Protocol accepts transmission from upper layer services (e.g.: packet data) and map them into fixed-sized ATM cells.
 - There are four types of data streams that are identified: Constant-bit rate, variable bit-rate, connection oriented packet data transfer, connectionless packet data transfer.

X.25

X.25 is an International Telecommunication Union-Telecommunication Standardization Sector (ITU-T) protocol standard for WAN communications that defines how connections between user devices and network devices are established and maintained. It is typically used in the packet-switched networks (PSNs) of common carriers, such as the telephone companies. Subscribers are charged based on their use of the network. The development of the X.25 standard was initiated by the common carriers in the 1970s. At that time, there was a need for WAN protocols capable of providing connectivity across public data networks (PDNs). X.25 is now administered as an international standard by the ITU-T.

X.25 Devices and Protocol Operation

X.25 network devices fall into three general categories: data terminal equipment (DTE), data circuit-terminating equipment (DCE), and packet-switching exchange (PSE). Data terminal equipment devices are end systems that communicate across the X.25 network. They are usually terminals, personal computers, or network hosts, and are located on the premises of individual subscribers. DCE devices are communications devices, such as modems and packet switches, that provide the interface between DTE devices and a PSE, and are generally located in the carrier's facilities. PSEs are switches that compose the bulk of the carrier's network. They

transfer data from one DTE device to another through the X.25 PSN. Figure 1.7 illustrates the relationships among the three types of X.25 network devices.

Figure below shows DTEs, DCEs, and PSEs that make up an X.25 Network

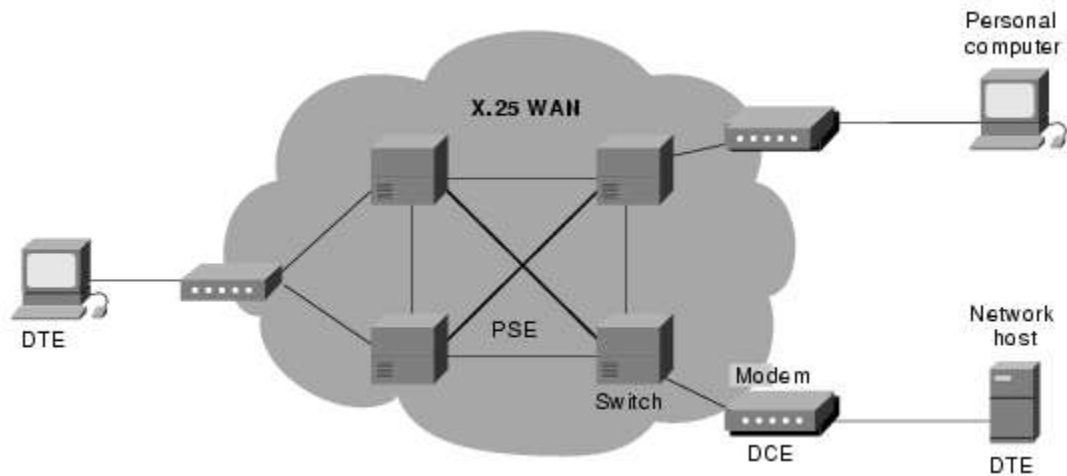


Figure: X.25 Network